

EXPERIMENT 6

Aim: Write a Program to implement Go-Back-N Sliding Window Protocol.

Solution:

Go-Back-N ARQ is a **sliding window error-control protocol** in which the sender can transmit multiple frames without waiting for an acknowledgment for each frame. If a frame is lost or damaged, the sender retransmits that frame and **all subsequent frames in the sender window** after a timeout.

Algorithm: Go-Back-N ARQ

1. **Start.**
2. **Read** the total number of frames (**n**).
3. **Read** the sender window size (**w**).
4. **Read** the frame number to be lost (**lostFrame**).
Enter -1 if no frame is to be lost.
5. **Initialize:**
 - base = 0 (first frame in the sender window)
 - nextFrame = 0 (next frame to be sent)
 - expected = 0 (next frame expected by the receiver)
6. **Repeat until all frames are transmitted** (base < n).
7. **Display the current sender window.**
8. **Send all frames in the current sender window one by one.**
9. **At the receiver:**
 - If the expected frame is **not the lost frame**:
 - Receive the frame.
 - Send an **ACK** for the received frame.
 - Increment expected.
 - Move the sender window forward by one position.
 - If there are more frames to send, send the next new frame.
 - Otherwise, if the expected frame **is lost**:
 - Display that the frame is lost.
 - Discard all subsequent **out-of-order frames**.
 - Display **Timeout Occurred**.
 - Retransmit the lost frame and all subsequent frames in the current sender window.
 - Receive the retransmitted lost frame.
 - Send its ACK.
 - Move the sender window forward.
 - Continue transmission.
10. **Repeat the process until all frames are successfully transmitted.**
11. **Display "All Frames Transmitted Successfully."**
12. **Stop.**

```
#include <stdio.h>
#include <conio.h>
int main()
{
    int n, w;
    int base = 0;
    int nextFrame = 0;
    int expected = 0;
```

```

int lostFrame;
int i, k;
clrscr();

printf("Enter total number of frames: ");
scanf("%d", &n);

printf("Enter sender window size: ");
scanf("%d", &w);

printf("Enter the frame number to be lost (-1 for no loss): ");
scanf("%d", &lostFrame);

printf("\n===== Go-Back-N ARQ =====\n");

while (base < n)
{
    /* Display Sender Window */
    printf("\nSender Window: ");
    for (i = base; i < base + w && i < n; i++)
        printf("%d ", i);
    printf("\n");

    /* Sender sends all frames in the current window */
    while (nextFrame < base + w && nextFrame < n)
    {
        printf("Sender --> Frame %d\n", nextFrame);
        nextFrame++;
    }

    /* Receiver processes frames */
    while (expected < nextFrame)
    {
        /* Frame Lost */
        if (expected == lostFrame)
        {
            printf("\n*** Frame %d Lost ***\n", expected);

            /* Receiver discards out-of-order frames */
            for (k = expected + 1; k < nextFrame; k++)
            {
                printf("Receiver: Frame %d Discarded (Out of Order)\n",
k);
            }

            printf("\nTimeout Occurred!\n");
            printf("Sender Retransmits from Frame %d\n\n", expected);

            /* Retransmit frames */
            nextFrame = expected;

            while (nextFrame < base + w && nextFrame < n)
            {
                printf("Sender --> Retransmit Frame %d\n", nextFrame);
                nextFrame++;
            }
        }
    }
}

```

```

    }

    /* Receiver now gets lost frame */
    printf("Receiver <-- Frame %d Received\n", expected);
    printf("Receiver --> ACK %d\n", expected);

    expected++;
    base++;

    /* Prevent same frame from being lost again */
    lostFrame = -1;

    break;
}
else
{
    printf("Receiver <-- Frame %d Received\n", expected);
    printf("Receiver --> ACK %d\n", expected);

    expected++;
    base++;

    /* Slide sender window */
    if (nextFrame < n)
    {
        printf("\nSender Window Slides\n");

        printf("New Sender Window: ");
        for (i = base; i < base + w && i < n; i++)
            printf("%d ", i);

        printf("\n");

        printf("Sender --> Frame %d\n", nextFrame);
        nextFrame++;
    }
}

}

printf("\nAll Frames Transmitted Successfully.\n");

return 0;
}

```

Output: